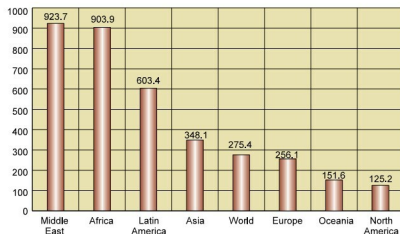


Growth Rate of Internet Users During 2000 to 2008

(in per cent)



Source: worldstats.com

Fig. 1: Growth in Internet usage

moves from one coverage or cell area to another, to ensure that he remains connected.

One simple solution is not to assign fixed IP address. There is a scheme to dynamically change the IP address by a protocol called dynamic host control protocol (DHCP). DHCP allots the intended user/computer a temporary IP address dynamically. It is a simple management solution.

But a temporary address has two major limitations: Other nodes cannot easily originate transmissions to the mobile node as they cannot effortlessly know about its new IP address, and each time a mobile computer obtains a new IP address, Internet client software applications must be restarted. Again, temporary addresses do not let the mobile users roam seamlessly from one IP subnet area to another. This is a handicap, especially for users in wireless local-area networks (LANs); for example, doctors with a wireless LAN connection need to maintain IP connections throughout the hospital as they access patient records. Here arises the need of a different solution—mobile IP.

Mobile IP enables mobile users to stay connected to the Internet regardless of their location and without

changing their IP address. It allows mobile users with a fixed IP address to connect to any IP subnet and immediately be reachable from the Internet. Mobile IP is basically an extension of the conventional IP that makes mobility transparent to higher layers of the Internet architecture.

A few definitions

IETF defines mobile computer as a device that users carry from one network to another while maintaining their permanent IP address. RFC (Request for Comments), 2002, defines mobile IP as “an enhancement of the Internet protocol that adds mechanisms for forwarding Internet traffic to mobile devices when they are connecting through other than their home network.” It is “an extension to the ‘untethered office’ concept, allowing unfettered access to the Internet or corporate networks any time, from any location.”

Currently, mobile IP is based on IPv4 (IP version 4) Internet. IP version 4 assumes that the IP address of a node (it is invariably also the IP address of the user or client) uniquely identifies its physical attachment to the subnet of Internet connectivity. When the source and destination nodes (such as a browser and a Web server) commu-

nicate over the Internet, these generally employ TCP for an end-to-end logical connection. When this logical connection is created, a TCP port number is assigned at both the source end and the destination end so that these two hosts remain connected.

Every end-to-end TCP logical connection is uniquely identified by four values: source IP address, source TCP port, destination IP address and destination TCP port. These values must be fixed and static for the duration of the connection.

One basic problem with mobile node is that it may be moving from network to network. Therefore the challenge in mobile IP is to manage the mobile node's IP address while it dynamically changes its location so that the mobile node always communicates with its permanent IP address.

Devices required for mobile IP

Fig. 3 shows the working principle of mobile IP. The devices required for mobile IP include the mobile node, home network, foreign network, care-of address and agents.

Mobile node. A node that roams and changes its position dynamically but keeps on communicating with any other system in the Internet as long as link-layer connectivity is available.

Home network. The subnet that the mobile node belongs to. Mobile node has its IP address with the home subnet. No mobile IP support is needed when the mobile node communicates within the home network.

Foreign network. The subnet, except home network. The current subnet is one which the mobile node is visiting now.

Care-of address. Mobile nodes are assigned two IP addresses: home address and care-of address.

The home IP address is a static address used to identify the end-to-end connection and used by the mobile node when connected to its home network.

The care-of address is not static. It